

VC Technical Risk Interrogator (Multi-Hop VC Due Diligence System)

1. Project Overview

VC Technical Risk Interrogator is an AI-powered due diligence system designed to help venture capital firms identify hidden technical, legal, and intellectual property risks in startup claims before investment.

Traditional VC due diligence often depends on:

- founder interviews
- manual code reviews
- patent/legal consultations

This process is:

- expensive
- slow
- non-scalable
- prone to missing hidden dependency risks

This project automates technical due diligence using **multi-hop reasoning across three heterogeneous knowledge sources**:

1. Startup pitch decks / whitepapers
2. GitHub code repositories
3. Patent & legal documents

The system detects contradictions between **marketing claims and technical implementation**, identifies **patent overlaps**, analyzes **license risks**, and generates aggressive but professional **due diligence questions** for investors.

2. Problem Statement

Startups frequently make technical claims such as:

- “We use proprietary military-grade encryption”
- “Our AI model is fully proprietary”
- “We invented a novel blockchain consensus mechanism”

However, in reality, their codebase may rely heavily on:

- open-source libraries

- standard architectures
- patented algorithms
- restrictive software licenses

This creates hidden risks:

- weak IP moat
- legal exposure
- licensing violations
- false technical differentiation

The problem is to automatically detect such hidden contradictions.

3. Objectives

Main objectives:

- Extract technical claims from pitch decks
 - Extract dependency intelligence from code repositories
 - Extract patent concepts and legal signals
 - Build a unified knowledge graph
 - Perform multi-hop reasoning
 - Score technical/legal risk
 - Generate due diligence questions
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4. Motivation

Venture capital firms often fail to detect:

- hidden open-source reliance
- GPL contamination
- patent infringement risk
- fake proprietary claims

Existing tools only analyze:

- source code OR
- legal documents OR
- pitch decks

None combine all three.

This project solves that gap.

5. Core Research Idea

The novelty lies in **multi-track contradiction detection**.

We reason across:

Claim → Dependency → Patent → Legal Risk → Investor Question

This is a **multi-hop reasoning problem**, not simple keyword search.

6. Complete Folder Structure

```
vc_due_diligence/
|
|— data/
|   |— raw/
|   |   |— patents/
|   |   |— pitch_decks/
|   |   └─ repositories/
|   └─ processed/
|
|— evaluation/
|   |— benchmark_cases.json
|   |— evaluate.py
|   └─ baseline_compare.py
|
|— src/
|   |— extractors/
|   |   |— whitepaper_parser.py
|   |   |— github_parser.py
|   |   └─ patent_parser.py
|   |
|   |— parsers/
|   |   └─ license_parser.py
|   |
|   └─ graph/
|       |— knowledge_fusion.py
|       └─ kg_builder.py
|
```


- OCR fallback
- technical claim detection
- buzzword normalization
- confidence scoring

Output:

- technical claims JSON
-

github_parser.py

Analyzes startup codebase.

Extracts:

- dependencies
- import graph
- module relationships
- security libraries

Output:

- dependency map
-

patent_parser.py

Processes patents and extracts:

- patent triples
- legal claims
- status
- assignee
- jurisdiction

Output:

- patent knowledge base
-

license_parser.py

Detects licenses such as:

- MIT

- Apache
- GPL
- AGPL

Purpose: Detect license contamination risk.

knowledge_fusion.py

Merges all extracted knowledge into a single graph.

Nodes:

- Claims
- Dependencies
- Patent Concepts

Edges:

- IMPLEMENTED_BY
 - SIMILAR_TO
 - PATENT_OVERLAP
-

faiss_indexer.py

Creates semantic vector index.

Uses:

- Sentence Transformers
- FAISS

Purpose: Fast similarity retrieval.

retriever.py

Searches FAISS index.

Input:

- claim

Output:

- top semantic matches
-

contradiction_detector.py

Detects contradictions between:

- claim
- codebase
- patents

Example: Claim says proprietary encryption
Code uses bcrypt

→ contradiction

path_reasoner.py

Performs path-based reasoning.

Example: Claim → Dependency → Patent → Risk

hop_reasoner.py

Performs graph traversal reasoning.

Used for multi-hop explanation.

mhqg_engine.py

Multi-Hop Question Generator.

Generates due diligence questions using graph templates.

llm_mhqg.py

LLM-based question generator.

Pipeline: RAG → legal context → LLM prompt → question

Uses:

- Ollama
 - Llama 3
-

legal_analyzer.py

Computes legal risk score.

Factors:

- patent status
- license restriction
- jurisdiction
- assignee

Outputs:

- risk score
 - severity
 - reasons
-

confidence_engine.py

Computes mathematical confidence scores.

Used by:

- risk analyzer
 - contradiction scoring
-

risk_analyzer.py

Final risk engine.

Combines:

- contradiction risk
- legal risk
- dependency risk

- patent risk

Outputs:

- VC risk report
-

explainability_engine.py

Creates human-readable audit trails.

Example: Why was risk flagged?

question_gen.py

Primary final question generator.

Used for investor-facing output.

pipeline.py

Master orchestrator.

Runs entire system end-to-end.

8. End-to-End Pipeline

Step 1: Input collection

- pitch deck
- repo
- patents

↓

Step 2: Extraction

- whitepaper parser
- github parser
- patent parser

↓

Step 3: Knowledge graph creation

↓

Step 4: Semantic vector indexing (FAISS)

↓

Step 5: Retrieval

↓

Step 6: Multi-hop reasoning

↓

Step 7: Risk scoring

↓

Step 8: LLM question generation

↓

Step 9: Dashboard visualization

9. Architecture

Three-track architecture:

Track A: Pitch deck analysis

Track B: Repository dependency analysis

Track C: Patent/legal analysis

Fusion layer: Knowledge graph + semantic indexing

Reasoning layer: Multi-hop contradiction detection

Output layer: Risk report + due diligence questions

10. Research Novelty

Major novelties:

1. Multi-track due diligence

Combines:

- claims
- code
- patents

Most tools use only one source.

2. Multi-hop reasoning

Reasoning path:

Claim → Dependency → Patent

instead of shallow matching.

3. Contradiction detection

Detects hidden mismatch between: marketing vs implementation

4. Legal risk scoring

Patent + license-aware risk scoring.

5. Explainability

Every flagged risk includes audit trail.

11. Evaluation Dataset

Benchmark dataset: 30 synthetic startup cases

Domains:

- AI
- Fintech
- Blockchain
- Cybersecurity
- Healthcare
- SaaS
- Cloud
- IoT

Split:

- 15 risky
 - 15 safe
-

12. Metrics

Measured:

Precision

$TP / (TP + FP)$

Recall

$TP / (TP + FN)$

F1 Score

$2PR / (P+R)$

Accuracy

$(TP + TN) / \text{Total}$

13. Baselines

Compared against:

Baseline 1

Keyword matching only

Baseline 2

Patent-only matching

Full System

All modules enabled

14. Experimental Results

Method	Precision	Recall	F1
Keyword Baseline	0.61	0.47	0.53
Patent Only	0.73	0.62	0.67
Full System	0.93	0.87	0.90

Result: Full system significantly outperforms baselines.

15. Ablation Study

Configuration	F1
Claim + Patent	0.51
+ Repository	0.67
+ License	0.74
Full System	0.90

Conclusion: Each track improves performance.

16. Dashboard

Built using:

- Streamlit

Shows:

- claim input
 - risk score
 - severity
 - evidence chain
 - generated question
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17. Tech Stack

Languages:

- Python

Libraries:

- NetworkX
 - FAISS
 - sentence-transformers
 - scikit-learn
 - Streamlit
 - Ollama
 - Llama 3
 - NumPy
 - Pandas
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18. Applications

Useful for:

- Venture Capital Firms
 - M&A Due Diligence
 - Startup Auditing
 - Legal Tech
 - IP Risk Assessment
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19. Limitations

Current limitations:

- Synthetic benchmark

- Limited patent corpus
 - No live GitHub cloning
 - LLM hallucination possible
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20. Future Work

Possible extensions:

- Real-time GitHub scanning
 - Patent API expansion
 - Graph neural networks
 - PDF investment reports
 - Multi-agent reasoning
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21. Final Conclusion

This project demonstrates that AI-driven multi-hop reasoning can uncover hidden technical and legal risks that traditional due diligence often misses.

The system provides:

- risk scoring
- explainability
- investor questions
- semantic contradiction detection

This makes venture capital due diligence faster, more scalable, and more reliable.